

Sales Performance and Product Demand Analysis for Improved Inventory Planning in a Retail Garments Store

A Final Report for the BDM Capstone Project

Submitted by:

Name: Sumaiya Afroz

Roll number: 24f1000754



IITM Online BS Degree Program,
Indian Institute of Technology, Madras,
Chennai Tamil Nadu, India, 600036

Content

1. Executive Summary	2
2. Detailed Explanation of the Analysis Process	2
2.1 Data Collection and Preprocessing	2
2.1.1 Data Collection	3
2.1.2 Data Cleaning	3
2.1.3 Feature Generation	3
2.2 Analytical Techniques	4
2.2.1 Descriptive Statistics (Daily & Weekly)	5
2.2.2 ABC Analysis	6
2.2.3 Inventory Turnover Analysis	6
2.2.4 Moving Average-based Demand Forecasting	7
2.3 Links to the Data and Analysis	8
3. Results and Findings	8
3.1 Results and Findings for Problem Statement 1: Reducing Locked Working Capital	8-9
3.1.1 ABC Analysis	9
3.1.2 Inventory Turnover Analysis	10-13
3.2 Results and Findings for Problem Statement 2	13
3.2.1 Descriptive Statistics	13-15
3.2.2 Moving Average Demand Forecasting	15-17
4. Interpretation of Results and Recommendations	17
4.1 Problem Statement 1: Reducing Locked Working Capital	17-18
4.2 Problem Statement 2: Minimising Total Inventory Cost	19
4.3 Overall Recommendation	20
5. Conclusion	21

1 Executive Summary

Matching Point is a B2C retail garment and matching accessories store located at Powerhouse Road, Maripur, Muzaffarpur, Bihar, owned by Md. Tarique Anwar. Established in 2014, the store sells sarees, suit pieces, dupattas, blouse materials, and related accessories but has relied entirely on manual, non-digitized records. This has caused two challenges: locked working capital due to slow-moving inventory, and high holding costs from demand-blind procurement.

Primary data covering 699 sales transactions across 56 products (January–April 2026) was collected through sales registers, bill books, and physical inventory checks, and organized into an Excel workbook with Item Master, Daily Sales, Inventory Summary, and Current Stock sheets. Descriptive statistics showed an average selling price of ₹417.45 and mean daily revenue of ₹7,435 (SD ₹3,497). The dataset was analyzed using Descriptive Statistics, ABC Analysis, Inventory Turnover Ratio, and Moving Average Forecasting in Microsoft Excel.

ABC Analysis (24 products, total revenue ₹9,07,565) showed that just 3 products—Suit Piece, Saree, and Dupatta—generate 67.56% of revenue, while 16 low-value products contribute only 10.14%. Turnover Analysis identified Night Wear, Men's Wear, and Accessories as the slowest-moving categories, showing where capital is locked, while Saree and Suit Piece turn over fastest. The 7-day Moving Average provided a smoother estimate of short-term demand than individual daily sales figures.

These findings show that reallocating capital from slow-moving Category C stock toward Category A fast-movers, combined with demand-based reorder points, can reduce locked capital and holding costs. The owner has shown interest in adopting the proposed ABC-based purchasing review and discount-driven clearance of slow stock once implemented, with detailed recommendations presented in Section 4.

2 Detailed Explanation of the Analysis Process

2.1 Data Collection and Preprocessing

2.1.1 Data Collection

Primary data was collected directly from Matching Point, a retail garment store in Muzaffarpur, Bihar, through field visits conducted during the January–April 2026 period. The data collection process focused on understanding the store’s sales patterns, inventory movement, and product-level performance. Sales-related information was obtained from the available handwritten sales records and customer bill books maintained by the business. Inventory-related information, including opening stock, quantity sold, and closing stock, was collected through physical stock verification and discussions with the business owner.

During the field visits, product names, sales quantities, selling prices, and inventory-related information were reviewed to understand the movement of different products. Discussions with the owner were also conducted to clarify product categories, stocking practices, and the way sales and inventory information was maintained. The collected information was subsequently organized into structured datasets for further analysis.

2.1.2 Data Cleaning

- The collected data was cleaned and standardized before performing the analysis to ensure consistency and reduce errors. The following cleaning and validation steps were performed:
- Incomplete records: 29 incomplete rows containing missing Date/Item ID information were identified in the raw `daily_sales_data` sheet and removed, leaving 699 valid transactions.
- Product standardization: Product names, categories, and sub-categories were standardized so that different naming variations of the same product were grouped under a single consistent label.
- Sales validation: Total sales values were verified using the relationship $\text{Total Sales} = \text{Quantity Sold} \times \text{Selling Price}$.
- Inventory consistency: Opening Stock, Quantity Sold, and Closing Stock were cross-checked using the relationship $\text{Closing Stock} = \text{Opening Stock} - \text{Quantity Sold}$.
- Inventory status validation: Inventory was classified into Normal, Near Stockout, and Overstock categories based on closing stock levels and sales movement.
- These steps helped ensure that the final dataset was consistent and suitable for product-level sales, ABC, inventory, and demand-related analysis.

2.1.3 Feature Generation

To support the inventory analysis, additional metrics were derived from the cleaned sales and inventory data.

These derived measures helped convert the available raw stock and sales information into indicators that could be used to evaluate inventory efficiency and product-level performance.

1 Average Inventory:

Average Inventory was calculated using the opening and closing inventory levels:

$$\text{Average Inventory} = (\text{Opening Stock} + \text{Closing Stock}) / 2$$

This measure provides an estimate of the inventory maintained during the analysis period and was used as the basis for evaluating inventory efficiency.

2 Inventory Turnover Ratio:

Inventory Turnover Ratio was used to measure the relative movement of inventory based on quantity sold:

$$\text{Inventory Turnover Ratio} = \text{Quantity Sold} / \text{Average Inventory}$$

A higher turnover indicates relatively faster movement of inventory, whereas a lower turnover indicates comparatively slower inventory movement. This metric was used alongside ABC analysis to identify products requiring greater inventory attention.

2.2 Analytical Techniques

2.2.1 Descriptive Statistics (Daily & Weekly)

Statistics	Daily Revenue (n=120 days)	Weekly Revenue (n=18 weeks)
Mean	₹7,435.13	₹49,567.50
Median	₹6,770.00	~₹48,113
Std. Deviation	₹3,497.00	₹13,560.06
Minimum	₹1,905.00	₹21,190.00
Maximum	₹19,360.00	₹74,005.00
Skewness	+0.96 (right-skewed)	-0.18 (near-symmetric)
Range	₹17,455.00	₹52,815.00
Total (Sum)	₹9,07,565	₹9,07,565

Table 1: Descriptive Statistics of Daily and Weekly Revenue

Method:

Table 1 shows Descriptive statistical analysis was used to summarize and understand the sales and demand patterns of the business. Measures such as mean, median, standard deviation, minimum, maximum, and skewness were calculated for daily and weekly revenue. These measures were used to examine the typical revenue level, variation in sales, range of revenue values, and the overall distribution of sales across different periods.

Calculation:

The descriptive statistics were calculated in Microsoft Excel using standard statistical functions. Mean was calculated as $\text{Mean} = \Sigma x/n$, while standard deviation was calculated as $s = \sqrt{[\Sigma(x - \bar{x})^2/(n - 1)]}$. Skewness was calculated to identify the direction and degree of asymmetry in the revenue distribution. In Excel, these measures were obtained using functions such as **AVERAGE**, **MEDIAN**, **STDEV.S**, **MIN**, **MAX**, and **SKEW**.

Justification:

Descriptive statistics was selected because it provides a clear quantitative summary of the business's sales performance and helps identify variations between different trading periods. Analysing measures such as average revenue and standard deviation helps determine the consistency of sales, while skewness provides an indication of whether sales are influenced by occasional high- or low-revenue periods. This analysis provides a statistical foundation for understanding demand patterns and supports better inventory and procurement planning. This is aligned with the project's objective of minimizing total inventory cost.

Visualization:

Daily and weekly revenue patterns were presented using appropriate tables and charts to make variations and trends easier to interpret.

2.2.2 ABC Analysis**Method:**

ABC analysis was used to classify products according to their contribution to total revenue. The products were first ranked in descending order of total revenue. The percentage contribution of each product and its cumulative revenue percentage were then calculated. Based on the cumulative contribution, products were classified into three categories: Class A (up to 70%), Class B (70–90%), and Class C (90–100%).

Formula:

- **Revenue Contribution (%) = (Product Revenue / Total Revenue) × 100**
- **Cumulative Revenue (%) = (Cumulative Revenue up to the Product / Total Revenue) × 100**

Justification:

ABC analysis was selected because the business does not currently use a structured quantitative approach to identify which products contribute most to revenue. The method helps distinguish high-revenue products from relatively low-contribution products and therefore supports prioritization of inventory and working capital. This is directly aligned with the first business problem of reducing locked working capital.

Visualization:

A Pareto chart was prepared to present product-wise revenue contribution together with the cumulative revenue percentage and the 70% and 90% classification boundaries.

2.2.3 Inventory Turnover Analysis**Method:**

Inventory turnover analysis was used to evaluate the rate at which inventory moves through the business. Average inventory was calculated using opening and closing stock, and the inventory turnover ratio was then calculated using quantity sold relative to average inventory.

Formula:

Average Inventory = (Opening Stock + Closing Stock) / 2

Inventory Turnover Ratio = Quantity Sold / Average Inventory

A higher turnover ratio indicates relatively faster inventory movement, whereas a lower turnover ratio indicates comparatively slower movement.

Justification:

Inventory turnover analysis was selected to identify products or product groups with slower stock movement that may keep working capital tied up in inventory for longer periods. This complements ABC analysis because revenue contribution alone does not indicate how efficiently inventory is moving. Together, the two methods help identify products that require closer inventory attention and support the objective of reducing locked working capital. This rationale is also consistent with the approved project methodology.

Visualization:

Inventory turnover values can be compared across product groups using a suitable comparative chart to identify relatively fast- and slow-moving inventory.

2.2.4 Moving Average-based Demand Forecasting**Method:**

Moving average forecasting was used to estimate near-future product demand from historical sales data. The method calculates the average demand over a fixed number of previous periods and uses this value as the forecast for the subsequent period. This helps smooth short-term fluctuations in sales and provides a simple estimate of the expected demand trend.

Formula:

$$\text{Moving Average Forecast} = (\text{Demand}_1 + \text{Demand}_2 + \dots + \text{Demand}_n) / n$$

where n represents the number of previous periods included in the moving average.

Justification:

Moving average forecasting was selected because the business currently relies largely on experience and observation for stocking and procurement decisions. The method provides a simple quantitative estimate of near-future demand using the sales records available from the business. By smoothing random fluctuations in transactional demand, it can support more structured procurement decisions and help reduce unnecessary overstocking and stockouts. This directly supports the second business problem of minimizing total inventory cost.

Visualization:

Historical demand and forecast values were presented through a line chart to make the underlying demand pattern and forecast trend easier to interpret.

2.3 Links to the Data and Analysis

1. Dataset and Complete Analysis (Excel Workbook): [View Dataset](#)
2. Evidence Folder (Photos, Interview, Authorization Letter): [View Evidence Folder](#)

Note: The linked workbook is best viewed in Google Sheets. Charts may render inconsistently when the file is downloaded and opened in Microsoft Excel due to differences in chart-rendering engines between the two platforms; the underlying data and formulas remain fully intact and verifiable in both formats.

3. Results and Findings

3.1 Results and Findings for Problem Statement 1: Reducing Locked Working Capital

3.1.1 ABC Analysis

The ABC analysis was conducted to identify the products that contribute most significantly to Matching Point's total revenue and to determine where inventory attention should be prioritized. The analysis covered 24 products with a total revenue of ₹9,07,565. The results show a strong concentration of revenue among a small number of products, indicating that all products should not be managed with the same level of inventory control.

Class	No. of Items	% of Items	Revenue (Rs.)	% of Revenue
A	3	12.50%	6,13,150	67.56%
B	5	20.83%	2,02,360	22.30%
C	16	66.67%	92,055	10.14%
Total	24	100%	9,07,565	100%

Table 2: ABC Classification Summary

Interpretation of ABC Classification Table

In Table 2, The ABC classification shows that the revenue contribution is highly concentrated among a small number of products. Class A consists of only 3 products (12.50% of the total products), but contributes 67.56% of total revenue, with Suit Piece, Saree, and Dupatta being the major contributors. Class B contains 5 products and contributes 22.30% of revenue, representing products with moderate financial importance. In contrast, Class C contains 16 products (66.67% of the product range) but contributes only 10.14% of total revenue. This indicates that inventory management should give greater attention to high-revenue Class A products, while the stock levels of the large number of Class C products should be reviewed carefully to reduce the possibility of unnecessary working capital being tied up in low-contribution inventory.

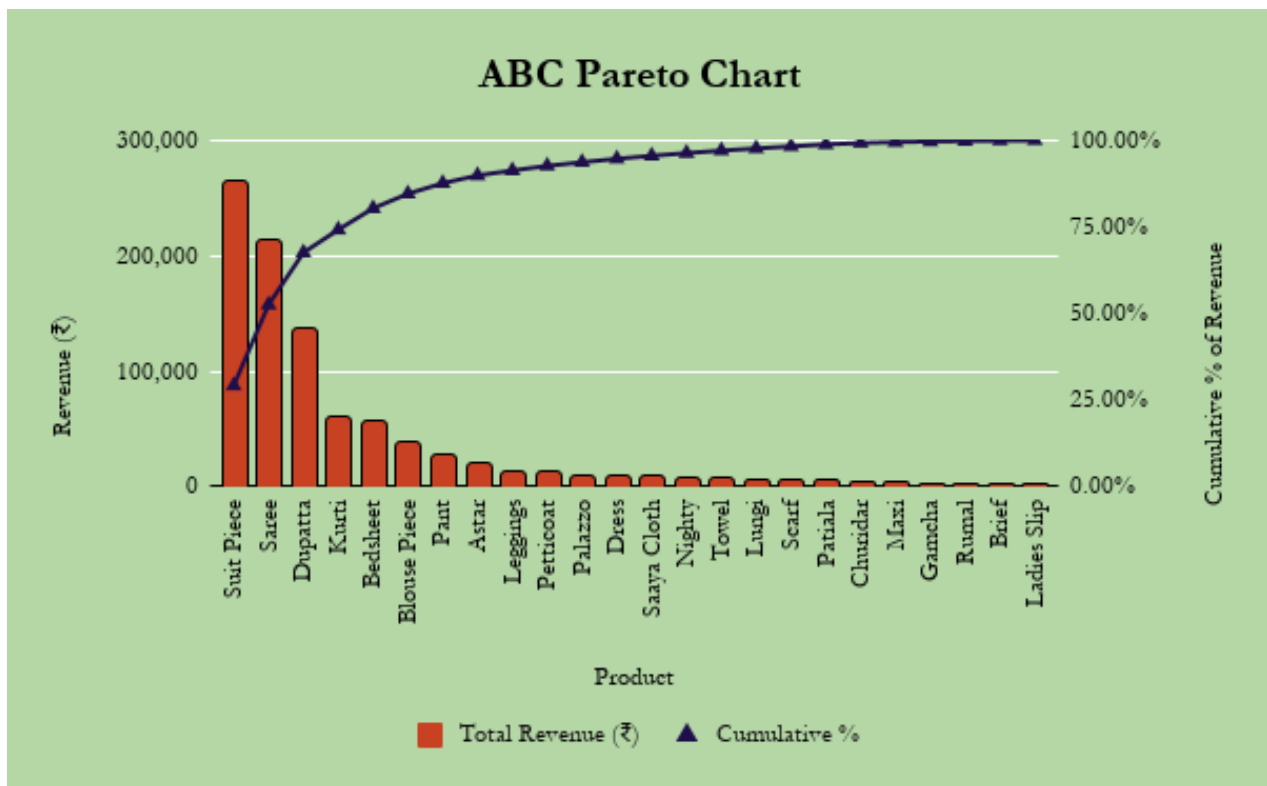


Figure 1: ABC Pareto Chart

In Figure 1, The Pareto chart provides a visual representation of the concentration of revenue across the 24 products. The bars are arranged in descending order of revenue, with Suit Piece showing the highest revenue contribution, followed by Saree and Dupatta. A sharp decline in the bar heights after the first few products indicates that revenue is concentrated among a limited number of products rather than being evenly distributed across the entire product range.

The cumulative revenue line rises rapidly across the first few products and reaches approximately 67.56% after the first three products, confirming that Suit Piece, Saree, and Dupatta together generate more than two-thirds of total revenue. The line then increases more gradually through the Class B products and eventually approaches 100% as the remaining Class C products are included.

This pattern provides a clear visual indication that the business has a relatively small group of high-impact products and a much larger group of low-revenue-contributing products. From an inventory-management perspective, this means that stock availability and replenishment decisions for the first few high-revenue products should receive greater attention. At the same time, the large number of Class C products should be reviewed carefully to avoid unnecessary accumulation of slow-moving inventory and consequent locking of working capital.

Therefore, the Pareto analysis supports a prioritized inventory-control approach rather than a uniform stocking strategy. The business can use the results to concentrate monitoring and procurement attention on high-revenue products while periodically reviewing the quantities and movement of lower-contribution products.

3.1.2 Inventory Turnover Analysis

1. Inventory Turnover Results

Category	Turnover
Saree	1.78
Bottom Wear	1.60
Suit Piece	1.60
Dupatta	1.53
Dress Material	1.52
Blouse Material	1.37
Ladies Wear	1.34
Matching Material	1.26
Home Textile	1.22
Undergarments	1.00
Accessories	0.90
Men's Wear	0.76
Night Wear	0.61

Table 3: Category-wise Inventory Turnover Ratio

2. Interpretation of Results

In Table 3, The inventory turnover results show considerable variation in stock movement across the product categories. Saree records the highest turnover ratio of 1.78, followed by Bottom Wear and Suit Piece at approximately 1.60, indicating comparatively faster movement of inventory in these categories. On the other hand, Night Wear records the lowest turnover of 0.61, followed by Men's Wear (0.76) and Accessories (0.90), indicating comparatively slower inventory movement. The difference suggests that inventory is not moving at the same rate across all categories, making a uniform stocking approach less suitable for the business.

3 Inventory Turnover Ratio by Category Chart

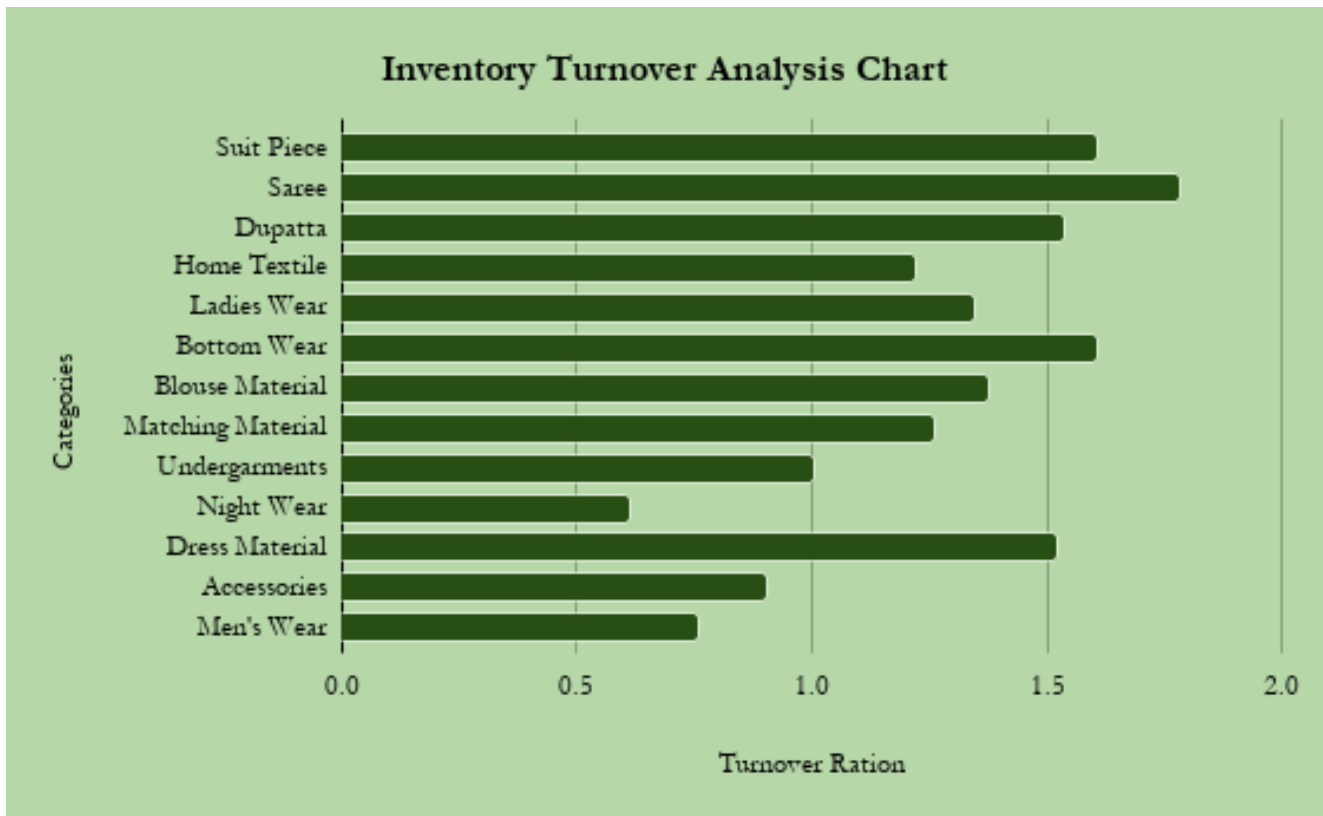


Figure 2: Inventory Turnover Ratio by Category

4 Chart Interpretation

In Figure 2, The chart makes the variation in inventory movement clearly visible. The relatively longer bars for Saree, Bottom Wear and Suit Piece indicate stronger stock movement, whereas the shorter bars for Night Wear, Men's Wear and Accessories highlight categories requiring closer inventory monitoring. This pattern is important because lower turnover means that inventory is being converted into sales comparatively slowly. If such categories are stocked in excessive quantities, the business may keep a larger amount of funds tied up in inventory for longer periods.

5 ABC × Turnover Risk Matrix

Category	Turnover Ratio	Revenue Level	Risk Status
Saree	1.78	High	Healthy - Maintain
Suit Piece	1.60	High	Healthy - Maintain
Bottom Wear	1.60	Medium	Healthy - Maintain
Dupatta	1.53	High	Healthy - Maintain
Dress Material	1.52	Low	Monitor
Blouse Material	1.37	Medium	Monitor
Ladies Wear	1.34	Medium	Monitor
Matching Material	1.26	Medium	Monitor
Home Textile	1.22	Medium	Monitor
Undergarments	1.00	Low	Review
Accessories	0.90	Low	Review
Men's Wear	0.76	Low	High Risk - Locked Capital
Night Wear	0.61	Low	High Risk - Locked Capital

Table 4: Combined ABC-Turnover Risk Matrix

Table 4 combines the revenue contribution level with the inventory turnover ratio for each category, providing a consolidated risk view rather than treating the two analyses independently. The matrix highlights categories requiring different levels of inventory attention based on revenue contribution and turnover.

However, Men's Wear and Night Wear emerge as the highest-risk categories — they combine low revenue contribution with the slowest turnover, indicating that capital invested in these categories is both underperforming and moving slowly, doubly justifying immediate corrective action such as discounting or reduced procurement.

6 Final Finding for Problem Statement 1

The inventory turnover analysis, when considered alongside the ABC analysis, indicates that inventory priority should be based on both revenue contribution and stock movement. High-performing categories need adequate availability to support sales, while low-turnover categories require controlled replenishment and periodic review. This can help Matching Point identify inventory that may remain unsold for longer periods and reduce the amount of working capital locked in slow-moving stock, directly addressing Problem Statement 1.

3.2 Results and Findings for Problem Statement 2

3.2.1 Descriptive Statistics

1. Results

Descriptive statistics were used to examine the level, variability, and distribution of daily and weekly sales revenue. The analysis helps identify periods of higher or lower sales and the extent of variation in revenue, providing a basis for inventory and procurement planning.

2. Interpretation of Results

The analysis indicates that demand is not evenly distributed across product categories. Some categories show comparatively higher demand, while others contribute less to overall demand. This variation suggests that applying the same stocking quantity across all categories may not be appropriate for the business.

The comparison also helps identify situations where closing inventory remains relatively high despite lower demand. Such a pattern can indicate a potential risk of overstocking, whereas categories showing stronger demand with relatively lower inventory may require closer monitoring to avoid stock shortages.

3 Demand vs Closing Inventory by Category



Figure 3: Demand vs Closing Inventory by Category

4. Chart Interpretation

Figure 3 highlights differences between demand and closing inventory across categories. Categories with higher demand generally require greater attention to stock availability, while categories with comparatively lower demand and relatively higher closing inventory may indicate excess stock. The variation between demand and remaining inventory shows that inventory levels are not always aligned with demand, which can contribute to higher holding costs and inefficient use of working capital.

5. Finding for Descriptive Statistics

Overall, the descriptive analysis indicates considerable variation in sales revenue across periods, suggesting that inventory decisions should account for changing demand patterns.. Identifying these differences provides a basis for improving stocking decisions and reducing the risk of both overstocking and understocking. This directly supports the objective of minimizing total inventory cost.

3.2.2 Moving Average Demand Forecasting

To estimate short-term demand and reduce the effect of day-to-day fluctuations in sales, a 7-day Moving Average technique was applied to the daily sales data of Matching Point. The moving average considers the sales of the recent seven days and calculates their average to provide a smoother representation of the underlying demand pattern.

Results of Moving Average Analysis

Parameter	Result
Forecasting Technique	7-Day Moving Average
Data Used	Daily Sales Data
Forecasting Window	7 Days
Purpose	Short-term demand estimation
Main Observation	Moving average smooths daily sales fluctuations

Table 5: Results of Moving Average Analysis

The analysis shows that actual daily sales fluctuate considerably from one day to another. For example, during the initial seven-day period, daily sales vary between ₹3,285 and ₹10,645, while the calculated 7-day moving average is approximately ₹7,541. This demonstrates how the moving average reduces the effect of unusually high or low sales on the overall demand estimate.

Graphical Analysis

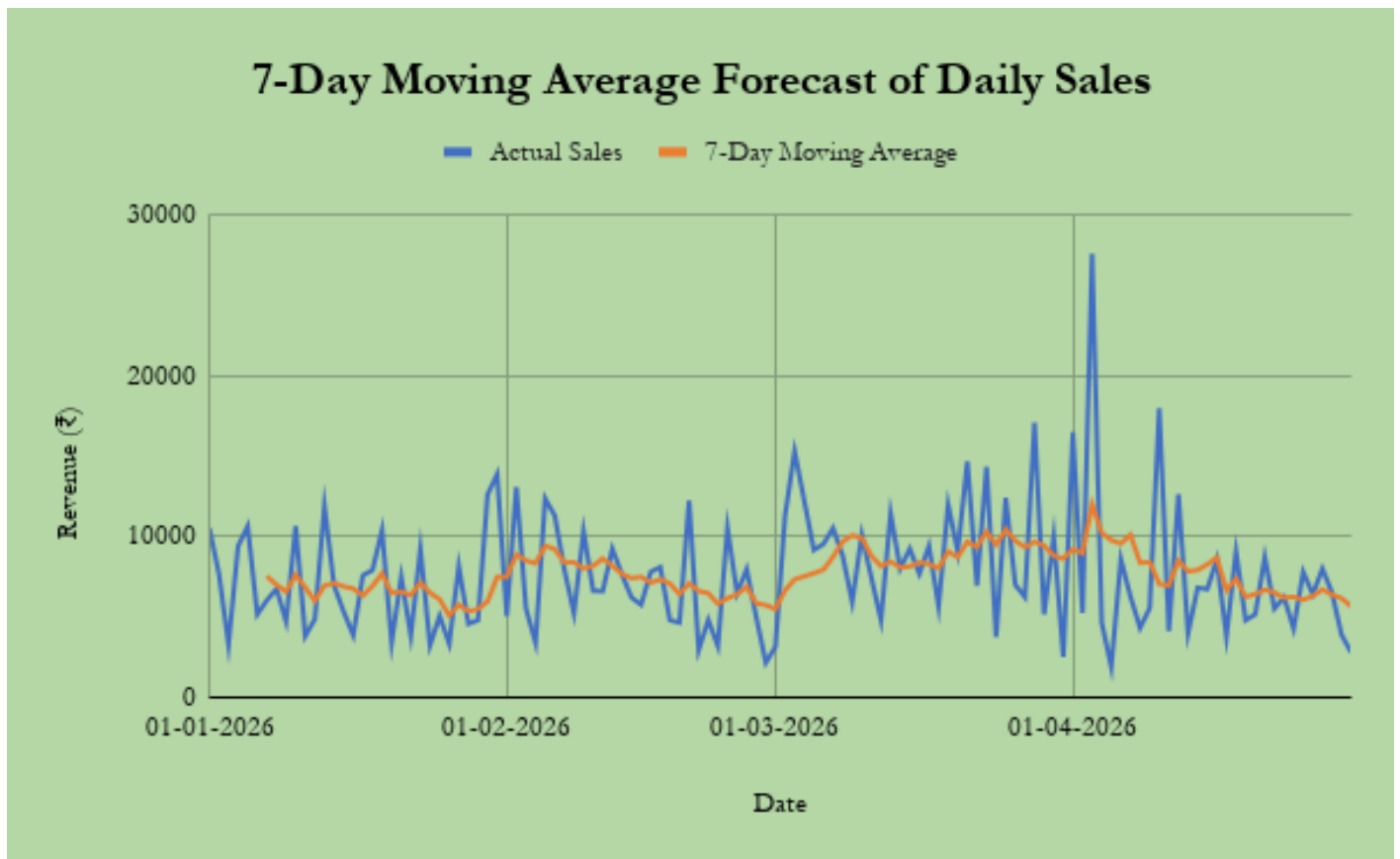


Figure:4 7-Day Moving Average Forecast of Daily Sales

Interpretation

In Figure 4, The graph clearly shows two different patterns. The Actual Sales line fluctuates sharply because customer purchases vary from day to day. There are several instances of sudden increases in sales as well as lower-sales days. In contrast, the 7-Day Moving Average line is considerably smoother because it averages the sales of the recent seven days.

The moving average therefore provides a clearer view of the underlying demand trend rather than focusing on individual daily fluctuations. The smoother line also indicates that short-term changes in sales do not necessarily represent a permanent change in customer demand.

For Matching Point, this analysis is useful because inventory decisions based only on a single day's sales may result in inappropriate purchasing decisions. A sudden high-sales day could lead to unnecessary over-purchasing, while a low-sales day could result in under-stocking. Using the recent seven-day average provides a more balanced reference for estimating near-term demand.

Key Finding

The 7-day moving average improves short-term demand visibility by smoothing irregular daily sales fluctuations. It can help Matching Point make more informed purchasing and inventory decisions based on recent demand patterns rather than relying entirely on intuition or individual-day sales.

This approach can contribute to better stock availability for regularly demanded products while reducing the possibility of unnecessary inventory accumulation, thereby supporting the project's objective of improving inventory planning and reducing working capital locked in stock.

4. Interpretation of Results and Recommendations

The analysis was conducted to understand the inventory and sales-related issues of Matching Point and to provide practical recommendations for improving inventory management. The findings from ABC analysis, inventory turnover analysis, demand versus closing inventory analysis, and moving average forecasting are interpreted below in the context of the two identified problem statements.

4.1 Problem Statement 1: Reducing Locked Working Capital

Interpretation of Results

The analysis shows that a significant portion of the business inventory is concentrated in a limited number of high-value and high-demand products. The ABC analysis indicates that the top three products—Suit Piece, Saree and Dupatta—contribute 67.56% of total revenue. This shows that these products have a major impact on the business revenue and therefore require closer inventory monitoring.

The inventory turnover analysis also shows differences in inventory movement across categories. Saree has the highest category-wise turnover ratio of approximately 1.78, while Night Wear has the lowest ratio of approximately 0.61. Lower turnover indicates comparatively slower movement of inventory, which can keep money tied up in stock for a longer period.

The current inventory analysis further shows that the business has 485 units of closing stock with approximately ₹1,02,600 of capital locked in inventory. Some products also have relatively high-value stock remaining. For example, the Work Suit category has a closing stock of 12 units at a cost price of ₹1,100 per unit, resulting in ₹13,200 of locked capital. This indicates that inventory value, and not only inventory quantity, should be considered while making purchasing decisions.

Overall, the findings suggest that the business can reduce locked working capital by giving greater attention to slow-moving and high-value inventory while ensuring adequate availability of high-demand products.

Recommendations

- **Prioritise high-value and high-demand products:**

Matching Point should maintain sufficient stock of high-revenue products such as Suit Piece, Saree, Dupatta and Kurti. The stock levels of these products should be reviewed weekly, and replenishment should be initiated whenever the available quantity approaches the defined minimum stock level.

- **Reduce slow-moving inventory:**

Products with low inventory turnover, particularly Night Wear and Men's Wear, should be reviewed monthly. If such products remain unsold for a prolonged period, the business should use discounts, bundle offers or promotional sales to clear excess stock and release locked working capital.

- **Use ABC classification for purchasing decisions:**

A-class products should receive the highest management attention, B-class products should be reviewed periodically, and C-class products should be purchased cautiously. This will help prevent excessive capital from being invested in low-contribution products.

Implementation and Expected Impact

Matching Point can implement these recommendations using the existing Excel-based inventory tracker without requiring expensive software. The owner can conduct a weekly stock review for high-demand products and a monthly review of slow-moving inventory. This will help reduce unnecessary stock accumulation, release working capital locked in slow-moving products, and maintain better availability of high-demand products.

4.2 Problem Statement 2: Minimising Total Inventory Cost

Interpretation of Results

The demand and closing inventory analysis indicates a mismatch between product demand and available stock for several categories. Total quantity sold during the analysis period was 2,172 units, compared with 485 units of closing stock. The category-wise comparison shows that some products have high demand but relatively low closing stock, increasing the possibility of stockouts.

The inventory status analysis provides further evidence of this imbalance. Out of 56 products, 24 products (42.86%) are classified as Near Stockout, 28 products (50%) are in Normal condition, while 4 products (7.14%) are classified as Overstock. Therefore, almost half of the tracked products require attention because their remaining stock is low.

For example, Dupatta recorded 553 units sold with 84 units of closing stock, while Saree recorded 244 units sold with only 15 units of closing stock. Similarly, Suit Piece recorded 210 units sold with only 26 units of closing stock. These results indicate strong demand and the need for timely replenishment.

The 7-day moving average analysis also shows that daily sales fluctuate considerably. The moving average smooths these fluctuations and provides a more stable estimate of underlying demand. At the end of April, the 7-day moving average was approximately 5,669 sales value per day, which can be used as a short-term reference for planning future stock requirements. These findings indicate that the business faces two opposite inventory risks: stockouts in high-demand products and excess stock in slower-moving products. Both situations can increase total inventory-related costs.

Recommendations

- **Introduce demand-based replenishment:**

Products frequently classified as Near Stockout should be reviewed weekly, and replenishment should be planned using recent sales and the 7-day moving average rather than relying on intuition.

- **Maintain minimum stock levels for high-demand products:**

Frequently sold products such as Dupatta, Saree, Suit Piece and other fast-moving items should have defined minimum stock levels. Reordering should be initiated whenever stock approaches the minimum level.

- **Control purchases of overstocked products:**

Products classified as Overstock should not be purchased again until existing stock reduces to an acceptable level. Where required, discounts or bundle offers can be used to improve their movement.

- **Maintain a monthly inventory review:**

At the end of every month, the business should compare quantity sold, closing stock and inventory status to identify products requiring replenishment or purchase reduction.

Implementation and Expected Impact

Matching Point can implement the proposed inventory system using a simple Excel-based tracker. The owner should update sales and stock information regularly, conduct weekly reviews of high-demand and Near Stockout products, and perform a monthly review of Overstock and slow-moving products. Using the 7-day moving average along with current stock status will support more informed purchasing decisions. The approach is expected to reduce unnecessary inventory accumulation, lower the risk of stockouts, release working capital locked in slow-moving products, and improve the balance between product availability and inventory cost.

4.3 Overall Recommendation

Based on the combined findings, Matching Point should move from intuition-based inventory decisions towards a simple data-driven inventory management system. The business should give priority to high-contribution products, monitor turnover and locked capital, use recent sales demand for replenishment, and regularly identify Near Stockout and Overstock products.

The proposed approach does not require expensive software and can be implemented through a structured Excel-based inventory system. Weekly monitoring of high-demand products, monthly review of slow-moving and overstocked products, and the use of recent sales trends for replenishment can help Matching Point gradually reduce locked working capital and unnecessary inventory costs while maintaining better availability of frequently demanded products.

5. Conclusion

This project set out to address two interconnected operational challenges at Matching Point: locked working capital arising from slow-moving inventory, and elevated inventory holding costs resulting from demand-blind procurement. Using 699 primary sales transactions collected over four months (January–April 2026), the study applied Descriptive Statistics, ABC Analysis, Inventory Turnover Ratio, and Moving Average Forecasting to move the business from an intuition-driven approach toward a structured, data-backed inventory management framework.

The findings establish that revenue at Matching Point is heavily concentrated: just 3 products (12.50%) account for 67.56% of total revenue, while 16 low-contribution products (66.67% of the range) generate only 10.14%. Inventory turnover analysis further revealed that categories such as Night Wear, Men's Wear, and Accessories move considerably slower than fast-moving categories like Saree and Suit Piece, indicating where working capital is currently tied up. The 7-day Moving Average provided a smoother basis for near-term procurement decisions than relying on single-day sales figures or owner intuition.

Collectively, these analyses indicate that a simple, low-cost Excel-based inventory tracking and review system—centered on ABC prioritization, turnover monitoring, and short-term demand forecasting—can help reduce locked working capital and minimise unnecessary inventory holding costs for Matching Point.